

2 Methodology

2.1 Research Design

This study contains a quantitative, cross-sectional design using secondary data obtained from Sri Lanka's Civil Registration and Vital Statistics (CRVS) system. In this study we observed about individual registered birth records in Sri Lanka, with parity or also known as birth order of each registered live births which is serving as primary outcome variable. Data from the years 2000,2005,2010,2015,2020 were used to examine patterns in parity and association of birth outcomes.

A cross-sectional approach is more suitable for this study because the data came from a single point in time within each selected year, and also the objective is to examine the distribution of parity and its associated maternal and birth related characteristics more than consider about changes over time. This design is consistent with practice in demographic and maternal health research using vital statistics and Demographic and Health Survey (DHS) data(Rutstein & Rojas, 2006; United Nations, 2014) . Cross-sectional designs are particularly well designed to population-based research,

The quantitative approach is more suitable because here the primary goal is to find out how birth order is distributed across Sri Lanka and identify which factors; such as maternal characteristics or social characteristics are strongly associated to families having more children. By using large, structured datasets and statistical regression models, study can account for many different variables at once. This method also satisfy that the study is transparent and can be easily replicated by other scientists using the same data and mathematical formulas.

The study is directed within the tradition of demographic and reproductive health research that uses administrative vital registration data to examine population level patterns of fertility and birth outcomes. This tradition is well established in the international literature, with comparable studies using CRVS data from the United Kingdom, Australia, the United States, and several South Asian nations (Bongaarts, 1998). Sri Lanka is an ideal setting for this type of analysis because it has rich birth registration and very detailed individual level records.

The study is explicitly descriptive and inferential rather than experimental. It does not involve any randomization, intervention, or manipulation of variables, and causal claims are accordingly not made. The regression models help to understand associations between predictors and parity, adjusted for the other variables included in each model, and these associations are interpreted with appropriate reference to the study's observational design.

2.2 Data Source and Study Population

The primary data source for this study is the birth registration data published by the Statistical Branch of the Registrar General's Department (No. 234/ A3 Denzil Kobbekaduwa Mawatha, Battaramulla) of Sri Lanka. Sri Lanka's Civil Registration and Vital Statistics system maintains nearly complete coverage of all live births in the country, making it a reliable population level data source for parity based research (United Nations, 2014).

The Sri Lankan CRVS system achieves high registration completeness relative to comparable middle income countries in South Asia. Estimated registration completeness for Sri Lanka in the early 2000s exceeds 97%, placing it among the best performing CRVS systems in the South and Southeast Asian region (UNICEF, 2019; United Nations, 2014) . This completeness is driven by some main factors such as high rates of hospital births facilitating

on site registration, legal deadlines for filing, and a comprehensive support network. By leveraging divisional secretariat offices and public health midwives, the system effectively reaches both rural and peri urban communities.

2.3 Dataset Structure and Scope

Table 2.1: Variables Available in the Birth Registration Dataset

Variable Name	Domain	Scale	Description
Birth_Order	Child	Ordinal	Birth order label: First through Ninth; primary outcome variable
Age of Mother	Maternal	Continuous	Maternal age at time of birth in completed years
Marital_Status	Maternal	Categorical	Marital status of the mother at time of birth
Race_of_Mother	Maternal	Categorical	Ethnic/racial group of the mother (Sinhalese, Tamil, Moor, Other)
Race_of_Father	Paternal	Categorical	Ethnic/racial group of the father
Gender	Child	Binary	Sex of the child (Male / Female)
Hospital or Not	Child/System	Binary	Whether the birth occurred in a hospital or other facility
Multiple_Birth_Status	Child	Categorical	Singleton, twin, triplet, or higher-order multiple birth
Birth_Weight (grams)	Child	Continuous	Birth weight of the child measured in grams
District_of_Mother	Geographic	Categorical	District of the mother's usual residence
Registered_District	Administrative	Categorical	District in which the birth was officially registered
Registered_Year / Month	Administrative	Discrete	Year and month in which the birth registration was recorded
Birth_Year / Month	Administrative	Discrete	Year and month in which the birth event occurred

The dataset is covering the entire country, with records contributed from all 25 administrative districts of Sri Lanka as they existed in 2000 to 2020 which are recently updated and published. The geographic context of the data supports sub national analysis, data integrity

and privacy are prioritized; all records are anonymized by the Registrar General's Department before release, following standard data-sharing protocols.

2.3 Description of Variables

2.3.1 Outcome Variable: Parity

Parity or Birth order, defined as the number of live births a woman has experienced which is the primary outcome variable of this study. In the raw dataset, birth order is recorded as text labels (First, Second, Third, Fourth, Fifth, Sixth, Seventh, Eighth, Ninth). For analytical purposes and for the fitting statistical models these labels were converted to their corresponding numeric values (1 through 9 respectively).

- Ordinal parity (1–9): used in the ordinal logistic regression model, preserving the full rank ordering of birth order.
- Count parity (1–9 as a count): used in the Poisson and Negative Binomial regression models, treating birth order as a cumulative count of live births.
- Binary parity: dichotomized as first-born (parity = 1, coded 1) versus later-born (parity ≥ 2 , coded 0), used in the binary logistic regression model to specifically address the characteristics of first births.

It is important to identify that this variable captures the birth order as recorded at the time of registration not the time when birth was occurred , based on information provided by the parents on the registration procedure because of this situation there can be births that may have occurred prior to the reference year but were not registered.

2.3.2 Explanatory Variables

Table 2.2: Operationalization of Explanatory Variables

Variable	Category	Groups / Scale	Reference Group
Age of Mother	Maternal	<20, 20–24, 25–29, 30–34, 35+	25–29 years (modal age group for first births)
Marital_Status	Maternal	Categories in registration data	First alphabetical category
Race_of_Mother	Maternal	Sinhalese, Tamil, Moor, Other	Sinhalese (majority group)
Gender	Child	Male / Female	Female
Hospital or Not	Child / System	Hospital / Not hospital	Not hospital
Multiple_Birth_Status	Child	Singleton / Multiple	Singleton
Birth_Weight (grams)	Child	Continuous (grams)	N/A — continuous predictor

2.3.2.1 Age of Mother

Maternal age is recorded in the data set as the age of mother when the birth was registered. This is a continuous variable and one of the most fundamental variables in parity-based studies. Usually maternal age lies between 15–49 its also known as reproductive age of women (WHO, 2006). In this study we consider about registered birth there for age of mothers can be out of those bounds because of registration delays. For regression modelling, maternal age was grouped into five categories: under 20 years, 20–24 years, 25–29 years , 30–34 years, and 35 years and above. This grouping is consistent with standard practice in demographic research using vital statistics (Rutstein & Rojas, 2006).

2.3.2.2 Marital Status

This variable carried out the legal marital status of mother at the time of birth registration. Marriage marks an important turning point in life, leading individuals to switch to new lifestyles that are closely related with reproductive and family related roles. Marital status is strongest determinants of fertility (Bongaarts, 1978). The CRVS system records marital status in the categories available in the registration form recognized under Sri Lanka personal law.

2.3.2.3 Race of Mother and Race of Father

Ethnicity is recorded in the CRVS system through separate variables for the mother and father as Race of Mother and Race of Father. As Sri Lanka is a multi-ethnic society, the CRVS system plays a crucial role in the legal recognition and administrative recording of these characteristics. The most common ethnic groups represented in the dataset are Sinhalese, Sri Lankan Tamil, Indian Tamil, Sri Lankan Moor (Muslim) , Burgher , and Malay.

2.3.2.4 Gender of child

The gender of the child represents the biological sex of the newborn, generally categorized as male or female, and is systematically recorded in the CRVS system as an important demographic characteristic used in population analysis. Gender is included as a predictor variable in parity based research due to the documented influence son or daughter preference on fertility behaviour (Bongaarts, 1998)

2.3.2.5 Place of Birth — Hospital or Not

The variable “Hospital or Not” is a binary indicator that records whether the birth took place in a hospital or other registered healthcare facility, or outside such a setting (e.g., at home or in a non-registered facility, state). This variable is important for analysis because the place of delivery may vary with parity. Mothers with higher parity may sometimes choose home births because of prior experience, while higher risk pregnancies may require more hospital care. Including this variable in the analysis helps to understand how the place of delivery is associated with birth order when other factors are considered.

2.3.2.6 Multiple Birth Status

The Multiple Birth Status variable indicates whether a registered birth is part of a singleton pregnancy or a multiple gestation (such as twin, triplet, or higher-order multiple births). In the dataset, the recorded categories include singleton, twin, triplet, and higher order multiples; however, the latter categories are rare and are often combined for analytical purposes. Multiple births are of particular demographic importance for several reasons. Multiple birth event simultaneously increases the recorded birth order for all children born in that event and also multiple births are associated with increased maternal and neonatal risks, and their occurrence is influenced by factors such as maternal age and the use of assisted reproductive technologies.

2.3.2.7 Birth Weight

Birth weight is an important sign of how the womb environment affected their growth before being born. Because of this, scientists have studied birth weight a lot to understand how it might relate to a person’s health; both right after birth and as they grow older into childhood

and adulthood. Birth weight is one of the most clinically significant indicators of neonatal health and a key determinant of infant mortality and long-term child development outcomes (Belbasis et al., 2016). In the raw data set weight is recorded as kilo grams and grams separately for the analysis purpose it changes into grams only.

2.3.3 Geographic and Administrative Variables

The dataset includes few geographic and administrative variables: District of Mother, Registered District, Registered Year/Month, and Birth Year/Month. These variables are supporting sub national analysis and the identification of temporal and administrative patterns in the data.

The District of Mother variable records the administrative district of the mother's usual place of living at the time of birth. Sri Lanka is divided into 25 administrative districts, grouped into nine provinces. District of residence is a meaningful geographic variable for demographic analysis because fertility rates, parity distributions, ethnicities, and access to healthcare services different across districts. The Registered District variable records the administrative district in which the birth was officially registered, which may differ from the mother's district of residence if the birth occurred in a hospital in a different district or if registration was completed at a different administrative office..

The Birth Year/Month and Registered Year/Month variables record the calendar year and month of the birth occurred and the birth registered, respectively. The difference between these two dates is the registration delay, usually registration must do within the three months after the birth occurred. Some times this registration delays cause for incomplete registration, and may be associated with specific demographic subgroups or geographic areas.

2.4 Data Preparation and Cleaning

Before statistical analysis, the dataset was carefully prepared to improve the quality of data, ensure consistency, and reduce errors and missing values. In this study data preparation process consist with some main steps;

1. Variable recording
2. Standardising categorical variables
3. Analysing and handling missing data
4. Checking outliers and valid ranges

Among these steps, outlier analysis and missing data handling are discussed in later sections. In the variable recoding stage, numerical codes of some specific variables in the raw dataset were mapped to their corresponding textual categories using predefined coding system. Microsoft Excel was used to implement and verify this mapping process, ensuring consistency and accuracy in the transformed variables.

Table 2.3: Registered/Birth month

Numeric code	Text Value
1	January
2	February
3	March
4	April
5	May
6	June
7	July
8	August
9	September
10	October

11	November
12	December

Table 2.4 Districts (before year 2002)

Numeric code	Text Value
1	Colombo
2	Gampaha
3	Kalutara
4	Kandy
5	Matale
6	Nuwara Eliya
7	Galle
8	Matara
9	Hambantota
10	Jaffna
11	Kilinochchi
12	Mannar
13	Vavuniya
14	Mullaitivu
15	Batticaloa
16	Ampara
17	Trincomalee
18	Kurunegala
19	Puttalam
20	Anuradhapura
21	Polonnaruwa
22	Badulla
23	Monaragala
24	Ratnapura
25	Kegalle
99	Nan

Table 2.5: Districts (after year 2002)

Numeric code	Text Value
11	Colombo
12	Gampaha
13	Kalutara
21	Kandy
22	Matale
23	Nuwara Eliya
31	Galle
32	Matara
33	Hambantota
41	Jaffna
42	Mannar
43	Vavuniya
44	Mullaitivu
45	Kilinochchi
51	Batticaloa
52	Ampara
53	Trincomalee
61	Kurunegala
62	Puttalam
71	Anuradhapura
72	Polonnaruwa
81	Badulla
82	Monaragala
91	Ratnapura
92	Kegalle
99	Nan

Table 2.6: Parity (birth order)

Numeric code	Text Value
1	First
2	Second
3	Third
4	Fourth
5	Fifth
6	Sixth
7	Seventh
8	Eighth
9	Nineth

Table 2.7: Gender

Numeric code	Text Value
1	Male
2	Female

Table 2.8 Marital status

Numeric code	Text Value
1	Married
2	Unmarried

Table 2.9: Birth place

Numeric code	Text Value
1	Hospital
2	Not in Hospital

Table 2.10: Multiple birth status

Numeric code	Text Value
1	Singelton
2	Twin
3	Triplets
4	Quadruplets
5	Quintuplets

Table 2.11: Race

Numeric code	Text Value
1	Sinhalese
2	Srilankan Tamil
3	Indian Tamil
4	Srilankan Moor
5	Burgher
6	Malay
7	Srilankan Chetty
8	Bharatha
9	Indian Moor
10	Pakistan Moor
11	Other Foreigners
12	Other Srilankan
13	NaN

2.4.1 Missing Variable Analysis

Missing data are common in administrative datasets such as the CRVS system, as the completeness of information depends on the reporting process and data availability at the time of registration. Therefore, analysis of missing data was conducted as a prior step. At each stage of data preparation, missing values were identified for all variables, and records containing any missing entries in key analytical variables were removed. This process resulted in a complete data set that was used for further analyses.

Missing values were identified for all variables, and the proportion of missing data was calculated for each variable as $(\text{number of records with missing value} / \text{total number of records}) \times 100\%$. Considering missingness, it has three main categories: low missingness ($\leq 2\%$), moderate missingness ($> 2\%$ to $\leq 10\%$), and high missingness ($> 10\%$). These categories follow the convention used in epidemiological research on missing data (Sterne et al., 2009; van Buuren, 2018).

In this study, birth order is the outcome variable. It was expected to have low missingness, as it is recorded at the time of birth. If records consist of missing birth order, those will be removed because they are inappropriate for our study. Then, maintain a complete data set that is suitable for our study.

2.4.2 Outlier Analysis

In this study, we mainly focused on two quantitative variables: Age of Mother and Birth Weight (grams). Outlier analysis was conducted for those two continuous variables because these are the only variables for which numerical extreme values can meaningfully occur and

require systematic assessment. Irregularities in categorical and discrete variables were addressed through the data cleaning and standardization procedures described earlier.

Basic statistical methods that used for outlier detection such as the interquartile range (IQR), z- score values and boxplot rules were not used as the primary methods for identifying valid values in this study because these variables carry clinically or demographically defined meanings, statistically derived thresholds may be scientifically inappropriate, because they treat the empirical distribution of the observed data as the reference standard rather than the biological or clinical knowledge that defines what values are possible and meaningful. In a large vital registration dataset such as the CRVS birth records, applying a z-score or IQR rule could be risk excluding biologically possible but rare observations (e.g., births to women aged 45–48, or very low birth weight infants) while potentially retaining because of data entry errors that happen to fall within the statistical bounds of the observed distribution.

Therefore for both variables , the valid range was defined using internationally recognized clinical and demographic standards. This approach follows established practice in parental epidemiology and demographic studies(Blencowe et al., 2019; Hughes et al., 2017; Rutstein & Rojas, 2006)

2.4.2.1 Outlier Analysis: Age of Mother

The standard demographic definition of female reproductive age, defined by World Health Organization (WHO) and United Nation (UN) Population Division, is 15 to 49 years. This age range is globally used in vital statistics reporting, DHS analyses, and global fertility monitoring and finding out eligible age group for reproductive health studies (Rutstein &

Rojas, 2006; United Nations, 2022; World Health Organization, 2023) The lower limit of 15 years represents the earliest age at which pregnancy is generally biologically possible, while the upper limit of 49 years reflects the decline and near cessation of fertility associated with late reproductive age.

In vital registration data, a small number of births may fall outside the standard reproductive age range. Maternal age under 15 years, although rare but biologically possible to occur and represent an important subgroup with specific health situation. Similarly, maternal aged 50 years and above are extremely uncommon but may occur during the perimenopausal period and are not necessarily errors. However, the CRVS system is also consist with data entry mistakes.

Based on this situation the valid range for maternal age was defined as 10 to 60 completed years. The lower bound of 10 years was chosen because, while extremely rare, pregnancy before age 15 is documented in the literature and in Sri Lankan vital statistics, and the WHO recognizes adolescent pregnancy from age 10 to 19 as a distinct reproductive health concern (World Health Organization, 2020) .The upper bound of 60 years was selected because natural conception after age 55 is so rare as to be clinically impossible without assisted reproduction, which was not widely used in Sri Lanka , therefore making any recorded age above 60 almost could be a data entry error rather than original maternal age. Records with ages in the ranges 10 to 14 and 50 to 60, while outside the standard WHO reproductive age range. The classification of age groups is in bellow table.

Table 2.12: Maternal Age Classification Applied in This Study

Age Range (years)	Classification	Action	Justification
< 10	Impossible- data entry error	Excluded	Below any recorded minimum age of menarche; biologically impossible
10 – 14	Adolescent pregnancy (below WHO standard window)	Retained; flagged	WHO recognizes pregnancy risk from age 10; rare but biologically possible (World Health Organization, 2020)
15 – 49	Standard WHO reproductive age window	Retained	UN / WHO global standard for female reproductive age (United Nations, 2022; World Health Organization, 2023)
50 – 60	Perimenopausal / menopausal pregnancy	Retained; flagged	Extremely rare but biologically documented; not necessarily a data error
> 60	Impossible- data entry error	Excluded	Natural conception after 60 clinically implausible without - Assisted Reproductive Technology; treated as entry error

2.4.2.2 Outlier Analysis: Birth Weight

The birth weight classifies into the following clinically recognized categories. Low birth weight (LBW) is defined as a birth weight below 2,500 grams, it is defined regardless of gestational age and has been used globally since the 1950s as the standard for identifying newborns at higher risk of illness and death (Hughes et al., 2017; World Health Organization, 2014). In the LBW category, very low birth weight (VLBW) is defined as below 1,500 grams, and extremely low birth weight (ELBW) as below 1,000 grams. Normal birth weight lies

between 2,500 to 3,999 grams, representing the range associated with the lowest risk of neonatal adverse outcomes. High birth weight, or macrosomia, is defined as 4,000 grams or more than that , with severe macrosomia typically set at 4,500 grams or above; macrosomia infants are at elevated risk of birth complications including shoulder dystocia and neonatal hypoglycemia (World Health Organization, 2004; Ye et al., 2015).

In addition to above categorize, biological limits of viability were used to define the outer bounds, with values outside this range treated as data entry errors. The minimum recorded weight of a surviving neonate recorded as approximately 220–250 grams, associated with extreme prematurity at the edge of viability; below approximately 300 grams, survival outside a highly specialized neonatal intensive care unit is not possible, and such care was not available in Sri Lanka for births of this weight (Stoll et al., 2015). At the upper extreme, birth weights above 6,000 grams are extremely rare, even in severe cases of macrosomia, and are therefore likely to represent data-entry errors in vital registration data (Hughes et al., 2017). Then the valid range for birth weight was set at 300 to 6,000 grams. Records outside this range were excluded from the analytical dataset. Observations in the VLBW ($<1,500$ g) and macrosomia ($\geq 4,000$ g) categories were retained but flagged for sensitivity analysis to assess their impact on regression estimates. This approach is consistent with best practice in perinatal research. (Blencowe et al., 2019).

Table 2.13: Birth Weight Classification Applied in This Study

Birth Weight Range	WHO Category	Action	Justification / Authority
< 300 g	Below viability threshold	Excluded	Below clinical viability; almost certainly a data-entry error (Hughes et al., 2017)
300 – 999 g	Extremely Low Birth Weight (ELBW)	Retained; flagged	WHO-defined category; viable only with specialist care (World Health Organization, 2004)
1,000 – 1,499 g	Very Low Birth Weight (VLBW)	Retained; flagged	WHO-defined category; requires intensive neonatal care (World Health Organization, 2004)
1,500 – 2,499 g	Low Birth Weight (LBW)	Retained	WHO global standard: LBW defined as < 2,500 g (World Health Organization, 2014)
2,500 – 3,999 g	Normal Birth Weight	Retained	WHO-defined normal range; lowest neonatal risk (World Health Organization, 2004)
4,000 – 5,999 g	Macrosomia / High Birth Weight	Retained; flagged	WHO/ACOG macrosomia threshold $\geq 4,000$ g (Ye et al., 2015)
> 6,000 g	Above plausible range	Excluded	Extremely rare; treated as data-entry error in CRVS context (Hughes et al., 2017)

2.5 Statistical Methods

2.5.1 Descriptive Statistics

In this study descriptive statistics were used to summarize the study population and examine the distribution of parity across maternal and child related characteristics. This stage serves two main purposes: providing complete idea of the dataset for users, and informing the specification of regression models, patterns of association and possible distributional issues that can be occurring in formal modeling. Frequency tables, graphical representations and percentage distributions were produced for all categorical variables, categorized by birth order. Bar charts and pie charts were used to present the overall distribution of the population in a simple manner, allowing readers with limited statistical knowledge to easily understand the patterns. Cross tabulation of parity against each predictor variables were computed to examine patterns of association before the stage of formal modeling. Boxplots, histograms were used to examine the distribution of the continuous predictor variables, but the validity of boxplot interpretations are described in early stage. For this analysis we will used data related to years 2000,2005,2010,2015 and 2020.

2.5.2 Chi-Square Tests of Association

Chi square test of independence was conducted to identify the bivariate association between birth order and each categorical predictor variable before move in to multivariate regression modelling. The chi square test evaluates whether the observed distribution of birth order differs across categories of a predictor more than would be expected under the null hypothesis of statistical independence (Agresti, 2018). The test statistic, degrees of freedom, and p-value were reported for each predictor.

A significance level noted as $\alpha = 0.05$ was applied for all hypothesis tests. Variables that showed a statistically significant association with parity in the bivariate chi square analysis were included as predictors in the multivariate regression models. This variable selection method is consistent with standard practice in demographic and health studies (Agresti, 2018; Hosmer et al., 2013). It is important to note that chi square tests are sensitive to sample size; in large datasets such as the CRVS data used here, even very small associations may produce statistically significant p-values.

Prior to regression modelling, the variance inflation factor (VIF) was computed for all predictor variables to compute multicollinearity. Variables with a VIF greater than 10.0 were considered for further examination for potential exclusion or combination with correlated predictors. VIF values between 5.0 and 10.0 were selected as moderate and reported with sensitivity analyses to assess their influence on model estimates.

2.5.3 Regression Model Selection

The birth order is an ordered variable ranging from 1 to 9, then three regression approaches were studied to address the research objectives. In these three models the outcome variable of this study recorded as different types summary of models given below table. For regression modeling we only consider about years 2010, 2015 and 2020.

Table 2.14: Summary of Regression Models Studied

Model	Outcome	Effect Measure	Justification
Ordinal Logistic Regression	Parity 1–9 (ordered)	Cumulative OR	Primary model; preserves full ordinal rank ordering; appropriate for ordered categorical outcomes
Poisson / Negative Binomial Regression	Parity 1–9 (count)	Incidence Rate Ratio (IRR)	Secondary; treats parity as count; grounded in fertility count regression literature
Binary Logistic Regression	Parity 1 vs ≥ 2 (binary)	Odds Ratio (OR)	Supplementary; addresses first-birth specific objective; dominant in South Asian parity studies

3.5.3.1 Model 1: Ordinal Logistic Regression (Proportional Odds Model)

Ordinal logistic regression, specifically the Proportional Odds Model (POM), was selected as the primary analytical model because this model more suitable when the outcome variable consists of ordered categories and the analysis aimed to preserve the rank ordering of birth order without collapsing categories into fewer groups (Hosmer et al., 2013). The POM estimates the cumulative odds of being at or below a given birth order category for a one unit increase in a predictor, holding all other variables constant.

The model is specified as:

The model is specified as:

$$\log \left(\frac{P(Y \leq j | X)}{P(Y > j | X)} \right) = \alpha_j - \beta X$$

where Y is the ordinal outcome (parity 1-9), j indexes the eight cumulative cut points, α_j are the category specific intercepts, X is the vector of predictors, and β is the vector of regression coefficients assumed constant across all cut points under the proportional odds assumption.

This approach is consistent with published parity research treating birth order as an ordinal variable (Das & Rahman, 2011).

A key assumption of the proportional odds model is the proportional odds (parallel regression) assumption, which states that the effect of each predictor remains constant across all cumulative cut points of the outcome variable. This assumption was tested by fitting separate binary logistic regression models at each of the eight cut points and examining the consistency of the estimated coefficients across these models. The Brant test was also applied as a formal statistical test. Any violations identified are reported in Chapter Four. Results are reported as odds ratios (OR) with 95% confidence intervals.

2.5.3.2 Model 2: Poisson and Negative Binomial Regression

As a complementary approach, birth order was treated as a count variable and modelled using count regression. This approach is grounded in the fertility literature, where Poisson regression is commonly used to model children ever born as a function of socio-demographic predictors. The standard Poisson regression model is specified as:

$$\log(\lambda) = \alpha + \beta_1 X_1 + \dots + \beta_k X_k$$

here λ represents the expected parity count. A key assumption of this model is Equi dispersion, meaning that the variance is equal to the mean. When overdispersion is present, Poisson regression produces underestimated standard errors and inflated test statistics. The choice between Poisson and Negative Binomial regression was therefore determined empirically by assessing overdispersion using the ratio of the variance to the mean of parity. . A dispersion ratio substantially greater than 1.0 warrants the use of Negative Binomial regression (Agresti, 2018). Fitted models were compared using AIC and BIC values. Results are reported as Incidence Rate Ratios (IRR) with 95% confidence intervals.

2.5.3.3 Model 3: Binary Logistic Regression

To directly address the first specific objective of the study mentioned as; examining the characteristics of first-born children relative to later-born children. The parity variable was recorded as first-born (parity = 1, coded 1) and later-born (parity ≥ 2 , coded 0). Binary logistic regression was then applied to identify predictors of first birth status. This approach is widely used in South Asian parity-based studies that distinguish between first and subsequent births (Adeleke, 2010; Hosmer et al., 2013). Model performance was evaluated using the Area Under the Receiver Operating Characteristic Curve. Classification accuracy at a 0.5 probability threshold was assessed using a confusion matrix. Results are reported as odds ratios with 95% confidence intervals.

2.6 Model Comparison and Selection

The three regression models were compared using standard information criteria. The AIC and BIC were computed for each model, with lower values indicating a better balance of fitted model. The log-likelihood was also reported for completeness. It is important to note that the three models are not directly comparable using information criteria because they operationalize the outcome variable differently. Direct comparison of AIC and BIC values across models with different outcome specifications is not strictly valid. Therefore, to get clear idea about those modes we have to focus on which approach is giving more meaningful explanation about the relationship between predictors and birth order. In addition to information criteria, the consistency of results across models was assessed.

2.7 Ethical Considerations

This study is based entirely on secondary, anonymized administrative data obtained from the Registrar General's Department of Sri Lanka. No primary data collection process involved and also this data set does not contain any personally identifiable information. As all records are anonymized releasing and publishing data can do under low legal situation.

2.8 Limitations of the Methodology

Several methodological limitations must be discussed. First, the cross-sectional design prevents causal inference. Associations between predictors and parity reflect statistical relationships at a single point in time and should not be interpreted as causal effects. Second, the dataset does not include variables such as maternal education, employment status, household income, which are recognized determinants of fertility behavior (Bongaarts, 1998). For example, the association between ethnicity and parity may partly reflect unmeasured educational differences across ethnic groups rather than a direct ethnic effect on fertility behavior. This limitation is inherent to CRVS data, which does not collect socioeconomic information at the point of birth registration. Third, while the CRVS system has high overall registration completeness, registration may be differentially incomplete for higher order births, remote geographic areas, or specific ethnic subgroups. Fourth, due to data availability and time constraints, regression model fitting was conducted using selected years (2010, 2015, and 2020). In contrast, descriptive analysis was performed using a broader set of years (2000, 2005, 2010, 2015, and 2020) to capture overall trends in parity at five-year intervals since 2000.